

Which Arab? How LLMs Treat Candidates from the 22 Arab League Countries in Hiring and Promotion

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Notes: title (not report text; rewrite in your own words)
Other title options, none of which assumes a result:

- Same CV, Different Passport: Auditing LLM Recruiters Across the Arab World
- One Label, 22 Nationalities: Do LLM Recruiters Tell Arab Applicants Apart?
- Is “Arab” One Category? A Nationality-Swap Audit of LLM Recruiters

Abstract

Notes: abstract (not report text; rewrite in your own words)

At most 250 words. The rubric asks for the research questions, the methods and the results. In all notes, (F) after a source means the full text was read and (A) means only the abstract was read: check the full text before citing a number from it.

- Context: LLMs are proposed for screening job applications. Audits so far mostly vary race and gender through names.
- Why Arab applicants: human employers in Western countries give about 41% fewer callbacks to Arab, North African and Middle Eastern names. LLMs repeat stereotypes about Arabs and Muslims in what they write. In hiring tasks the evidence is thin and mixed. Numbers and sources: introduction notes.
- Why inside the Arab world: nationality openly sorts workers there, and LLM recruiting tools are being built for these markets.
- Gap: every study treats “Arab” as one group. To our knowledge, none lets an LLM decide on applicants while single Arab nationalities are varied.
- Questions (from the proposal; hypotheses are not written yet): (1) With the CV held fixed, do a model’s ratings and choices differ across the 22 Arab League nationalities, and by more than across unrelated nationalities? (2) Local versus foreign applicants. (3) Does a paragraph telling the model to ignore nationality change this? (4) Do hiring and promotion, and the four models, agree?
- Method: 20 synthetic jobs with one CV each; every job placed in each of the 22 Arab League countries; one Nationality: line with 29 conditions; rate one applicant (score 0–100, interview yes/no); choose between two applicants with the same CV; four open-weight models.

- Results: nothing has been run as of 2026-10-04. Write this part last, and only from the blocks that were run.
- Last sentence: what follows. The data decide: either the single category hides differences, or it is adequate for these models.

1 Introduction

Notes: motivation, step by step (not report text; rewrite in your own words)

- **1. LLMs in hiring.** Many audits of LLMs that screen CVs; almost all vary race and gender through names. Nationality is rare: it appears in 13.2% of 189 papers on LLM bias, gender in 79.9% (not specific to hiring). Sources: gaebler2024auditing (F), armstrong2024silicon (F), an2025measuring (A), ghosh2025bias (A).
- **2. Human employers.** Meta-analysis of field experiments: callback ratio 0.59 for Arab, Maghrebi and Middle Eastern applicants, about 41% fewer positive callbacks, from 31 effects. Higher after 2000 than in the 1990s in six Western countries. France: 22.8% vs 33.3% callbacks (9,600 applications; a report, not a journal article). Netherlands: 31% vs 46%. Australia: 22% vs 35%. Sweden, Iraq-born: 10% vs 21%. Sources: lippens2023state (F), quillian2023trends (F), arnault2021discrimination (F), thijssen2021ethnic (F), booth2012does (F), vernby2019immigrants (F).
- **3. LLMs.** In generated text: Muslims tied to violence in 66% of GPT-3 completions; Arabs rather than Westerners named the “loser group” in 79% of model-by-topic comparisons (the prompt forces a choice); MENA names 78% under-represented in generated workplace stories. In hiring tasks: Arab names 1.41 points lower on a 1–100 scale in GPT-3.5 (selection ratio 0.85 at a cut-off); GPT-4 sends Arabic names to the lower-status job more often; no effect in Gemini 2.0 Flash; some newer models favour the minority. Sources: abid2021persistent (A), saeed2024desert (A), shieh2026intersectional (F), lippens2024computer (F), bai2025explicitly (A), hoffmann2026evaluating (F), bunel2026customer (F), arcuschin2026blind (F).
- **4. One category.** All of these treat “Arab” as one group. A name cannot do better: raters assigned 34% of Moroccan names to Morocco. Among human employers, origin can matter (Germany: Iraqi and Moroccan below Turkish; Sweden: Iraq-born 10%,

Somalia-born 5%). Inside the Arab world the little evidence points the other way: no significant nationality differences among Qatari citizens; Arab expatriates in Kuwait and Qatar treat fellow Arabs as one group. Sources: martiniello2022signaling (F), koopmans2019taste (F), vernby2019immigrants (F), shockley2024sharing (F), blaydes2023arab (F).

- **5. Why the Arab world.** Nationality openly sorts workers there. Qatar, low-income workers: Egyptian QR1,454 vs Nepali QR853 a month (plain averages). Jordan: segmentation “based not on education or initial skills but rather on nationality”. Gulf states expelled workers by Arab nationality in 1990–91. An LLM recruiting pipeline exists for the Saudi market; it codes nationality as Arab vs non-Arab. Sources: gardner2013portrait (F), razzaz2017challenging (F, ILO report), kapiszewski2006arab (F, UN paper), albaroudi2026addressing (F).
- **6. Gap and what we do.** No study was found in which an LLM decides on applicants while single Arab nationalities are varied (searches of 29–30.09 and 04.10.2026; research/novelty_assessment.md). Write “to our knowledge”. Then the question and one sentence on the design.

Notes: research questions (as in the proposal of 2026-10-04) (not report text; rewrite in your own words)

- Framing: the 22 countries differ in wealth, stability and how their citizens are seen abroad (high-income Gulf states; middle-income Egypt, Jordan, Morocco; countries marked by war and displacement such as Syria, Yemen, Sudan, Libya, Somalia). Does a model carry that into decisions about individuals?
- RQ1 (main): do scores and decisions differ across the 22 Arab League nationalities with everything else identical, and more than across the placebo nationalities?
- RQ2: do the differences follow a country’s economic and political position (income group, armed conflict)? The conflict classification is not fixed yet.
- RQ3: does the place of the job matter: local versus foreign, and the same foreign nationality in a Gulf country versus a North African one?
- RQ4: do the differences remain with the ignore-nationality paragraph, and when a lottery is allowed?
- RQ5: same patterns in hiring and promotion, and across models?

Notes: why the scenario is realistic (not report text; rewrite in your own words)

- Many Arabs work in another Arab country. Gulf states: Arabs from other countries were 72% of the foreign population in 1975, 56% in 1985 and 32% in 2002/4. Jordan: Jordanian, Egyptian and Syrian workers in one labour market. Sources: kapiszewski2006arab (F, UN paper), razzaz2017challenging (F, ILO report).
- Politics has singled out single Arab nationalities: 1990–91, more than 1.5 million people expelled from Gulf states, among them up to one million Yemenis from Saudi Arabia, 200,000 Jordanians and 150,000 Palestinians. Source: kapiszewski2006arab (F).

- Models hold country-specific associations: Libyan, Sudanese and Tunisian among the five most negative demonyms in GPT-2 text; GPT models suggest additional scrutiny for applicants from Iraq, Sudan and Somalia in simulated immigration decisions; models represent the values of some Arab countries much better than others. Sources: venkit2023nationality (F), mao2025gatekeepers (F), zahraei2025menavalues (F).
- No source yet for Arabs working in Egypt or in North African countries. Find one before writing that.

Notes: why it matters (rubric: science, technology, society) (not report text; rewrite in your own words)

- Science: bias audits report one number for “Arab”. If single nationalities differ, that number hides it. If they do not, pooling is justified for these models. Either result answers the question.
- Technology: LLM screening tools are proposed for Arab labour markets (albaroudi2026addressing, F). How they react to a nationality line is untested.
- Society: nationality already sorts workers in these markets (step 5). A screener that ranks nationalities would repeat that at scale. This last sentence is our own argument and has no source.

Notes: do not overstate (not report text; rewrite in your own words)

- Not “LLMs discriminate against Arab applicants” as a fact: one small penalty, one null, some reversals.
- The callback numbers are from Western countries. No field experiment with applicant nationality was found inside an Arab country.
- Not “Arab nationalities are treated differently in Arab countries”: that is barely studied. Say that nationality matters in these labour markets.
- Say “Arab League nationalities”, not “MENA” and not “Arab applicants”. MENA in other studies includes Turkey or Iran.
- The claim to be first is tentative. One paper is still unread in full (bilon2025sociodemographic).
- Do not cite numbers from mohammad2026mirage: they are placeholders.
- This is an audit of model behaviour, not a hiring tool. The score is not a hiring probability.

2 Related Work

Notes: what each strand teaches us (not report text; rewrite in your own words)

The rubric asks for what is relevant to our work, not for a list of papers.

- **LLM hiring audits.** Mostly US race and gender through names (gaebler2024auditing, F; armstrong2024silicon, F). The direction is unstable: in one study only the 2023 model favours White applicants and every later model shows no gap or the reverse (gao2026can, A); in another, every detected race or ethnicity bias favours the minority (arcuschin2026blind, F). For us: expect either direction.
- **Order and ties.** The first-listed candidate is chosen 63.5% of the time (rozado2026gender, F): we show each pair in both orders. With identical CVs, models mostly call a tie when they may (at least 94% ties for 7 of 9 models), and forbidding ties creates apparent gaps (chen2026beyond, F; vohra2026audit, F). This is why experiment 2 has a lottery version and a must-choose version, and why the must-choose version needs care.
- **Explicit cues.** A stated attribute moves decisions more than a name (tamkin2023evaluating, F). Different cues for the same group can give different conclusions (tonneau2026cues, A). For us: the results hold for a nationality line, not for names.
- **Arab group in LLM studies.** Always pooled: lippens2024computer (F), hoffmann2026evaluating (F), bai2025explicitly (A), albaroudi2026addressing (F). Stereotypes in text: abid2021persistent (A), saeed2024desert (A), shieh2026intersectional (F).
- **Nationality in LLM decisions.** 20 nationalities in job recommendations, Jordan the only Arab one; the condition without a nationality behaved like an outlier (salinas2023unequal, F). For us: the CV without a nationality line is not a neutral zero. GPT-4 prefers developers from some countries, tested by swapping only the location string (nakano2024nigerian, F).
- **Telling the model to ignore the attribute.** Brought discrimination close to zero in Claude 2.0 (tamkin2023evaluating, F). The best instruction cut the gap between groups by 27% on average (huijzer2025discrimination, F). Instructions fail once realistic company context is added (karvonen2025robustly, F). For us: experiment 3 can go either way.
- **Human evidence.** See the introduction notes, steps 2, 4 and 5.
- **Promotion.** No source on promotion decisions has been searched yet. Find one, or justify the scenario in a sentence.

3 Data

Notes: materials (not report text; rewrite in your own words)

- 20 job ads, one per occupation: 7 low-skill, 6 mid, 2 mid-high, 5 high. Employer names are invented. Every ad requires English only.
- 20 CVs, one per job. Middle qualification level: meets the requirements, not outstanding. No names of places, schools or companies.
- Setting: every job is set once in each of the 22 Arab League countries.

- Nationality: one line in the CV. 29 conditions: 22 Arab League nationalities, 2 benchmarks (German, Turkish), 4 placebo nationalities (Uruguayan, Malawian, Maldivian, Cambodian), 1 without a nationality line.
- Everything is synthetic: no real applicant, CV or employer.
- Still needed for the rubric: descriptive statistics and plots of the answers.

Source: the README files under experiment/.

Notes: why the CVs carry no names (not report text; rewrite in your own words)

The CV shows a reference number instead of a name. Example of a CV header:

Applicant reference: APP-4106
Location: Jordan
Nationality: Egyptian

- A name cannot tell Arab countries apart. It signals “Arab” as one group only; the nationality line can signal each of the 22.
- A name adds signals we do not want: gender, religion, sometimes region.
- The same name would be on every version of a CV, also next to a non-Arab nationality, which sends a mixed signal.
- A missing name is not neutral either, so the main prompt tells the model that applications are pseudonymised.

Sources in custom.bib: armstrong2024silicon, chen2026beyond, gaebler2024auditing, krause2012anonymous (only the abstract was read).

Notes: nationalities whose Arab identity is contested (not report text; rewrite in your own words)

Comoros, Djibouti and Somalia are Arab League members whose Arab identity is contested. They stay in every comparison. Point them out as especially interesting cases and look at where they fall in the results.

4 Methods

Notes: experiment setup (not report text; rewrite in your own words)

- One run: a main prompt (the model is an HR recruiter for the company in the country), then a follow-up message with the job ad and the CV(s).
- Experiment 1, rate one person: fit score 0–100 and yes/no.
- Experiment 2, choose between two people with the same CV and different nationalities. Two versions: A, B or lottery; A or B.
- Experiment 3: experiments 1 and 2 again, with one added paragraph in the main prompt telling the model to ignore nationality.
- Two scenarios: hiring (interview) and promotion.
- Three wordings of every follow-up message, used in rotation.

- Pairs for experiment 2: per job and country, the 22 Arab nationalities are randomly paired into 11 pairs; each pair is shown in both orders.
- Models: Qwen3 8B, Llama 3.1 8B, Gemma 3 12B, Mistral Small 3.2 24B. State the exact versions and settings.
- How the answers are analysed: not decided as of 2026-10-04.

Planned runs per model:

Block	Runs
1 Hiring, rate one	12,760
2 Hiring, choose between two	19,360
3 Blocks 1 and 2 with ignore paragraph	32,120
4 Promotion, everything	64,240
Total	128,480

Source: experiment/README.md.

5 Results

Notes: nothing to report yet (not report text; rewrite in your own words)

As of 2026-10-04 no run exists. The proposal lists what to compare:

- the spread of scores and interview rates across the 22 Arab League nationalities, against the spread across the 4 placebo nationalities;
- which nationality is chosen in experiment 2, how often the model takes the lottery, and what it does when the lottery is removed;
- local versus foreign applicants;
- with versus without the ignore paragraph;
- whether the four models agree.

The hypotheses and the analysis have to be written down before the first real run. Block 1 on all four models is the plan for the midterm.

6 Conclusion and Discussion

Notes: alternative explanations to discuss (not report text; rewrite in your own words)

- A nationality line also says “local” or “foreigner”. Every job is run in every country, so the two can be separated: compare the local nationality with the others. In human hiring, lacking citizenship costs callbacks (quillian2026racialized, A).
- The label may stand for religion, conflict or the income of the country, not for the nationality as such. country_covariates.csv has income and region.
- Any set of labels spreads the scores a little. The placebo nationalities are the yardstick; with only four of them it is a rough one.
- Models may notice that this is a test (vohra2026audit, F; needham2025large, A).

- One run per combination and no repeats: the chance variation of a single answer is not measured on its own.
- A result of “no difference” is an answer too. It would fit the little human evidence from inside the Arab world (shockley2024sharing, F; blaydes2023arab, F).

Limitations

Notes: not the full benchmark (not report text; rewrite in your own words)

- The full plan is 128,480 runs per model for four models. Because of hardware and time limits, not all of it was run.
- FILL IN AFTER THE RUNS: which blocks and which models were actually run, and what that means for the conclusions. As of 2026-10-04 nothing has been run.
- Other limits to consider: one CV per job; synthetic CVs and job ads; English only; nationality given as one explicit line; open models of 8 to 24 billion parameters only.

Author Contributions

Use of AI Tools

Notes: what to declare (not report text; rewrite in your own words)

As of 2026-10-04, assisted by Claude Code: literature search and research notes, design documents, the prompts, the job ads and CVs, the first draft of the project proposal, repository code and structure, title suggestions, and these notes (including the notes for the abstract, introduction and related work). Keep the list up to date.

Declaration

A Prompts

Notes: prompts (not report text; rewrite in your own words)

Copy in, unchanged, the main prompts and one follow-up message per experiment from experiment/prompts/.